

AARON MUELLER

CONTACT	Center for Computing and Data Sciences (CDS) Boston University 665 Commonwealth Ave., Office 806 Boston, MA 02215 (USA)	amueller@bu.edu aaronmueller.github.io
RESEARCH INTERESTS	• Natural language processing • Mechanistic and causal interpretability • Causality • Computational psycholinguistics	
EDUCATION	Johns Hopkins University Ph.D., Computer Science M.S.E., Computer Science GPA: 3.9/4.0 <i>Thesis:</i> Emergent Syntactic Behaviors and Mechanisms in Neural Language Models <i>Advisors:</i> Tal Linzen, Mark Dredze	Baltimore, MD Aug. 2023 May 2020
	New York University Visiting academic, Center for Data Science	New York, NY Aug. 2021 – Aug. 2023
	University of Kentucky B.S., Computer Science. <i>Honors.</i> B.S., Linguistics. <i>Honors.</i> GPA: 4.0/4.0. <i>Summa cum laude.</i>	Lexington, KY May 2018 May 2018
ACADEMIC POSITIONS	Boston University <i>Assistant Professor</i> , Department of Computer Science	Boston, MA Jul. 2025 – Present
	Northeastern University <i>Zuckerman Postdoctoral Fellow</i> , Khoury College of Computer Sciences <i>Advisor:</i> David Bau	Boston, MA Aug. 2023 – Jun. 2025
	Technion – Israel Institute of Technology <i>Zuckerman Postdoctoral Fellow</i> , Department of Computer Science <i>Advisor:</i> Yonatan Belinkov	Haifa, Israel Aug. 2023 – Jun. 2025
INDUSTRY EXPERIENCE	Meta <i>Research Intern</i> <i>Manager:</i> Kanika Narang – Research in retrieval-augmented generative models for few-shot question answering. – Resulted in improved F_1 on multiple QA and classification datasets using far fewer parameters than state-of-the-art models. Also resulted in a publication at ACL [C28].	Menlo Park, CA May – Nov. 2022
	Amazon Web Services (AWS) <i>Applied Scientist Intern</i> <i>Manager:</i> Saab Mansour – Research in pre-training methods for improving goal-oriented dialogue agents. – Resulted in state-of-the-art few-shot intent classification accuracy (>30% 1-shot gains) and a publication at ACL [C33].	Santa Clara, CA May – Aug. 2021

Raytheon BBN Technologies

Research Intern

Manager: Ilana Heintz

Cambridge, MA

May – Aug. 2019

- Research in low-resource cross-lingual word alignment and entity linking.
- Implemented convolutional neural machine translation models rivaling our prior seq2seq model’s BLEU with over 20% faster training and over 50% faster inference.

PUBLICATIONS Peer-reviewed Conference Articles

- C1. Sankaran Vaidyanathan, David Arbour, **Aaron Mueller**, Scott Niekum, David Jensen. “**The Curse of Multiple Mediators: Hidden Interaction Effects in Activation Patching.**” To appear in *Proceedings of the Fourtieth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
- C2. Kevin Lu, Jannik Brinkmann, Stefan Huber, **Aaron Mueller**, Yonatan Belinkov, David Bau, Chris Wendler. “**Mechanisms of AI Protein Folding in ESMFold.**” To appear in *Proceedings of the Fourtieth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
- C3. Jacob Brinton, Jannik Brinkmann, Mark Crovella, **Aaron Mueller**. “**The Interlingua Hypothesis: LLMs Translate via a Latent Task-agnostic Feature Space.**” Accepted to the *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.
- C4. Keren Fuentes, **Aaron Mueller**. “**Who Do Language Models Think Is Competent? A Mechanistic Analysis of Occupational Bias.**” To appear in *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.
- C5. Kerem Sahin, Sheridan Feucht, Adam Belfki, Jannik Brinkmann, **Aaron Mueller**, David Bau, Chris Wendler. “**In-context Learning Beyond Copying: A Training-time Analysis of Abstractive ICL.**” To appear in *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.
- C6. **Aaron Mueller**, Andrew Lee, Shruti Joshi, Ekdeep Singh Lubana, Dhanya Sridhar, Patrik Reizinger. “**From Isolation to Entanglement: When Do Interpretability Methods Identify and Disentangle Known Concepts?**” In *Proceedings of the Association for Computational Linguistics (ACL)*, 2026.
- C7. John Seon Keun Yi, **Aaron Mueller**, Dokyun Lee. “**Latent Agents: A Post-Training Procedure for Internalized Multi-Agent Debate.**” In *Proceedings of the Association for Computational Linguistics (ACL)*, 2026.
- C8. Deniz Bayazit, **Aaron Mueller**, Antoine Bosselut. “**Crosscoding Through Time: Tracking Emergence & Consolidation of Linguistic Representations Throughout LLM Pretraining.**” In *Proceedings of the Association for Computational Linguistics (ACL)*, 2026.
- C9. Tomer Ashuach, Dana Arad, **Aaron Mueller**, Martin Tutek, Yonatan Belinkov. “**CRISP: Persistent Concept Unlearning via Sparse Autoencoders.**” In *Proceedings of the Association for Computational Linguistics (ACL)*, 2026.
- C10. Zilu Tang, Qiao Zhao, Gabriel Franco, Derry Wijaya, **Aaron Mueller**, Sebastian Schuster, Najoung Kim. “**Do Language Models Track Entities Across State Changes?**” In *Proceedings of the International Conference on Machine Learning (ICML)*, 2026.
- C11. Shruti Joshi, **Aaron Mueller**, David Klindt, Wieland Brendel, Patrik Reizinger, Dhanya Sridhar. “**Causality is Key for Interpretability Claims to Generalise.**” In *Proceedings of the International Conference on Machine Learning (ICML)*, 2026.
- C12. Shengao Wang, Wenqi Wang, Zecheng Wang, Max Whitton, Michael Wakeham, Arjun Chandra, Joey Huang, Pengyue Zhu, Helen Chen, David Li, Jeffrey Li, Shawn Li, Andrew Zagula, Amy Zhao, Andrew Zhu, Sayaka Nakamura, Yuki Yamamoto, Jerry Jun Yokono, **Aaron Mueller**, Bryan A. Plummer, Kate Saenko, Venkatesh Saligrama, Boqing Gong. “**BabyVLM-**

V2: Toward Developmentally Grounded Pretraining and Benchmarking of Vision Foundation Models.” In *Proceedings of the Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

- C13. Ekdeep Singh Lubana, Can Rager, Sai Sumedh R. Hindupur, Valerie Costa, Greta Tuckute, Oam Patel, Sonia Krishna Murthy, Thomas Fel, Daniel Wurgaft, Eric J. Bigelow, Johnny Lin, Demba Ba, Martin Wattenberg, Fernanda Viegas, Melanie Weber, **Aaron Mueller**. “**Priors in Time: Missing Inductive Biases for Language Model Interpretability**.” In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026.
- C14. Zhengyang Shan, **Aaron Mueller**. “**Measuring Mechanistic Independence: Can Bias Be Removed Without Erasing Demographics?**” In *Proceedings of the European Chapter of the Association for Computational Linguistics (EACL)*, 2026.
- C15. Joe Stacey, Lisa Alazraki, Aran Ubhi, Beyza Ermis, **Aaron Mueller**, Marek Rei. “**Improving the OOD Performance of Closed-Source LLMs on NLI Through Strategic Data Selection**.” In *Proceedings of the European Chapter of the Association for Computational Linguistics (EACL)*, 2026.
- C16. Dana Arad, **Aaron Mueller**, Yonatan Belinkov. “**SAEs Are Good for Steering – If You Select the Right Features**.” In *Proceedings of the Annual Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.
- C17. **Aaron Mueller***, Atticus Geiger*, Sarah Wiegrefe, Dana Arad, Iván Arcuschin, Adam Belfki, Yik Siu Chan, Jaden Fiotto-Kaufman, Tal Haklay, Michael Hanna, Jing Huang, Rohan Gupta, Yaniv Nikankin, Hadas Orgad, Nikhil Prakash, Anja Reusch, Aruna Sankaranarayanan, Shun Shao, Alessandro Stolfo, Martin Tutek, Amir Zur, David Bau, Yonatan Belinkov. “**MIB: A Mechanistic Interpretability Benchmark**.” In *Proceedings of the International Conference on Machine Learning (ICML)*, 2025. [*Equal contribution]
- C18. Tal Haklay, Hadas Orgad, David Bau, **Aaron Mueller**, Yonatan Belinkov. “**Position-aware Automatic Circuit Discovery**.” In *Proceedings of the Annual Conference of the Association for Computational Linguistics (ACL)*, 2025.
- C19. Michael Hanna*, **Aaron Mueller***. “**Incremental Sentence Processing Mechanisms in Autoregressive Transformer Language Models**.” In *Proceedings of the Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*, 2025. [*Equal contribution]
- C20. Jannik Brinkmann, Chris Wendler, Christian Bartelt, **Aaron Mueller**. “**Large Language Models Share Representations of Latent Grammatical Concepts Across Typologically Diverse Languages**.” In *Proceedings of the Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*, 2025.
- C21. Juan Diego Rodriguez, **Aaron Mueller**, Kanishka Misra. “**Characterizing the Role of Similarity in the Property Inferences of Language Models**.” In *Findings of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*, 2025.
- C22. Samuel Marks, Can Rager, Eric J. Michaud, Yonatan Belinkov, David Bau, **Aaron Mueller**. “**Sparse Feature Circuits: Discovering and Editing Interpretable Causal Graphs in Language Models**.” In *International Conference on Learning Representations (ICLR)*, 2025. **Oral Presentation (top 1.8%)**.
- C23. Jaden Fiotto-Kaufman, Alexander R. Loftus, Eric Todd, Jannik Brinkmann, Caden Juang, Koyena Pal, Can Rager, **Aaron Mueller**, Samuel Marks, Arnab Sen Sharma, Francesca Lucchetti, Michael Ripa, Adam Belfki, Nikhil Prakash, Sumeet Multani, Carla Brodley, Arjun Guha, Jonathan Bell, Byron Wallace, David Bau. “**NNsight and NDIF: Democratizing Access to Foundation Model Internals**.” In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.

- C24. Yaniv Nikankin, Anja Reusch, **Aaron Mueller**, Yonatan Belinkov. “**Arithmetic Without Algorithms: Language Models Solve Math With a Bag of Heuristics.**” In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.
- C25. **Aaron Mueller**, Albert Webson, Jackson Petty, Tal Linzen. “**In-context Learning Generalizes, But Not Always Robustly: The Case of Syntax.**” *Proceedings of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.
- C26. Eric Todd, Millicent L. Li, Arnab Sen Sharma, **Aaron Mueller**, Byron C. Wallace, David Bau. “**Function Vectors in Large Language Models.**” In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.
- C27. **Aaron Mueller**, Tal Linzen. “**How to Plant Trees in Language Models: Data and Architectural Effects on the Emergence of Syntactic Inductive Biases.**” In *Proceedings of the Association for Computational Linguistics (ACL)*, 2023.
- C28. **Aaron Mueller**, Kanika Narang, Lambert Mathias, Qifan Wang, Hamed Firooz. “**Meta-training with Demonstration Retrieval for Efficient Few-shot Learning.**” In *Findings of the Association for Computational Linguistics (ACL)*, 2023.
- C29. Alex Warstadt*, **Aaron Mueller***, Leshem Choshen, Ethan Wilcox, Juan Ciro, Rafael Mosquera, Bhargavi Paranjabe, Adina Williams, Tal Linzen, Ryan Cotterell. “**Findings of the BabyLM Challenge: Sample-efficient Pretraining on Developmentally Plausible Corpora.**” In *Proceedings of the BabyLM Challenge at the Conference on Computational Natural Language Learning (CoNLL)*, 2023. [*Equal contribution]
- C30. Koustuv Sinha, Jon Gauthier, **Aaron Mueller**, Kanishka Misra, Keren Fuentes, Roger Levy, Adina Williams. “**Language Model Acceptability Judgements Are Not Always Robust to Context.**” In *Proceedings of the Association for Computational Linguistics (ACL)*, 2023. **Out-standing Paper Award.**
- C31. Julian Michael, Ari Holtzman, Alicia Parrish, **Aaron Mueller**, Alex Wang, Angelica Chen, Divyam Madaan, Nikita Nangia, Richard Yuanzhe Pang, Jason Phang, Samuel R. Bowman. “**What Do NLP Researchers Believe? Results of the NLP Community Metasurvey.**” In *Proceedings of the Association for Computational Linguistics (ACL)*, 2023.
- C32. **Aaron Mueller**, Robert Frank, Tal Linzen, Luheng Wang, Sebastian Schuster. “**Coloring the Blank Slate: Pre-training Imparts a Hierarchical Inductive Bias to Sequence-to-sequence Models.**” In *Findings of the Association for Computational Linguistics (ACL)*, 2022.
- C33. **Aaron Mueller**, Jason Krone, Salvatore Romeo, Saab Mansour, Elman Mansimov, Yi Zhang, Dan Roth. “**Label Semantic Aware Pre-training for Few-shot Text Classification.**” In *Proceedings of the Association for Computational Linguistics (ACL)*, 2022.
- C34. **Aaron Mueller**, Yu Xia, Tal Linzen. “**Causal Analysis of Syntactic Agreement Neurons in Multilingual Language Models.**” In *Proceedings of the Conference on Computational Natural Language Learning (CoNLL)*, 2022.
- C35. Alexandra DeLucia, Shijie Wu, **Aaron Mueller**, Carlos Aguirre, Mark Dredze, Philip Resnik. “**BERNICE: A Multilingual Pre-trained Encoder for Twitter.**” In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022.
- C36. **Aaron Mueller**, Mark Dredze. “**Fine-tuning Encoders for Improved Monolingual and Zero-shot Polylingual Neural Topic Modeling.**” In *Proceedings of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2021.
- C37. **Aaron Mueller**, Zach Wood-Doughty, Silvio Amir, Mark Dredze, Alicia L. Nobles. “**Demographic Representation and Collective Storytelling in the Me Too Twitter Hashtag Activism Movement.**” In *Proceedings of the Association for Computing Machinery (ACM) on Human-Computer Interaction (HCI), Computer-Supported Cooperative Work and Social Computing (CSCW)*, 2021.

- C38. Matthew Finlayson*, **Aaron Mueller***, Sebastian Gehrmann, Stuart Shieber, Tal Linzen, Yonatan Belinkov. “**Causal Analysis of Syntactic Agreement Mechanisms in Neural Language Models.**” In *Proceedings of the Association for Computational Linguistics (ACL)*, 2021. [*Equal contribution]
- C39. **Aaron Mueller**, Garrett Nicolai, Panayiota Petrou-Zeniou, Natalia Talmina, Tal Linzen. “**Cross-linguistic Syntactic Evaluation of Word Prediction Models.**” In *Proceedings of the Association for Computational Linguistics (ACL)*, 2020.
- C40. **Aaron Mueller**, Garrett Nicolai, Arya D. McCarthy, Dylan Lewis, Winston Wu, David Yarowsky. “**An Analysis of Massively Multilingual Neural Machine Translation for Low-Resource Languages.**” In *Proceedings of the Language Resources and Evaluation Conference (LREC)*, 2020.
- C41. Arya D. McCarthy, Rachel Wicks, Dylan Lewis, **Aaron Mueller**, Winston Wu, Oliver Adams, Garrett Nicolai, Matt Post, David Yarowsky. “**The Johns Hopkins University Bible Corpus: 1600+ Tongues for Typological Exploration.**” In *Proceedings of the Language Resources and Evaluation Conference (LREC)*, 2020.
- C42. Garrett Nicolai, Dylan Lewis, Arya D. McCarthy, **Aaron Mueller**, Winston Wu, David Yarowsky. “**Fine-grained Morphosyntactic Analysis and Generation Tools for More Than One Thousand Languages.**” In *Proceedings of the Language Resources and Evaluation Conference (LREC)*, 2020.
- C43. Marten van Schijndel, **Aaron Mueller**, Tal Linzen. “**Quantity Doesn’t Buy Quality Syntax with Neural Language Models.**” In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019.
- C44. Arya D. McCarthy, Winston Wu, **Aaron Mueller**, Bill Watson, David Yarowsky. “**Modeling Color Terminology Across Thousands of Languages.**” In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019.

Peer-reviewed Journal Articles

- J1. **Aaron Mueller**, Jannik Brinkmann, Millicent Li, Samuel Marks, Koyena Pal, Nikhil Prakash, Can Rager, Aruna Sankaranarayanan, Arnab Sen Sharma, Jiuding Sun, Eric Todd, David Bau, Yonatan Belinkov. “**The Quest for the Right Mediator: Surveying Mechanistic Interpretability for NLP Through the Lens of Causal Mediation Analysis.**” *Computational Linguistics*, 2026.
- J2. Ethan Gotlieb Wilcox, Michael Hu, **Aaron Mueller**, Tal Linzen, Alex Warstadt, Leshem Choshen, Chengxu Zhuang, Ryan Cotterell, Adina Williams. “**Bigger Is Not Always Better: The Importance of Human-scale Language Modeling for Psycholinguistics.**” In *Journal of Memory and Language (JML)* 144, 2025.
- J3. Ian R. McKenzie, Alexander Lyzhov, Michael Martin Pieler, Alicia Parrish, **Aaron Mueller**, Ameya Prabhu, Euan McLean, Xudong Shen, Joe Cavanagh, Andrew George Gritsevskiy, Derik Kauffman, Aaron T. Kirtland, Zhengping Zhou, Yuhui Zhang, Sicong Huang, Daniel Wurgaft, Max Weiss, Alexis Ross, Gabriel Recchia, Alisa Liu, Jiacheng Liu, Tom Tseng, Tomasz Korbak, Najoung Kim, Samuel R. Bowman, Ethan Perez. “**Inverse Scaling: When Bigger Isn’t Better.**” In *Transactions on Machine Learning Research (TMLR)*, 2023. **Featured Paper.**

Peer-reviewed Workshop Articles

- W1. Tal Haklay, Nikhil Prakash, Sana Pandey, Antonio Torralba, **Aaron Mueller**, Jacob Andreas, Tamar Rott Shaham, Yonatan Belinkov. “**Pitfalls in Evaluating Interpretability Agents.**” To appear in *Proceedings of the BlackboxNLP Workshop at the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.
- W2. **Aaron Mueller**. “**Missed Causes and Ambiguous Effects: Counterfactuals Pose Challenges for Interpreting Neural Networks.**” *The Mechanistic Interpretability Workshop at the International Conference on Machine Learning (ICML)*, 2024. **Honorable Mention for Top Paper.**

- W3. Alina Klerings, Christian Bartelt, **Aaron Mueller**. “**Developmentally Plausible Multimodal Language Models Are Highly Modular.**” In *Proceedings of the BabyLM Challenge at the Conference on Computational Natural Language Learning (CoNLL)*, 2024.
- W4. Aruna Sankaranarayanan, Dylan Hadfield-Menell, **Aaron Mueller**. “**Disjoint Processing Mechanisms of Hierarchical and Linear Grammars in Large Language Models.**” In *ICML Workshop on Large Language Models and Cognition*, 2024.
- W5. Alexandra DeLucia*, **Aaron Mueller***, Xiang Lisa Li, João Sedoc. “**Decoding Methods for Neural Narrative Generation.**” In *Proceedings of the Workshop on Generation Evaluation and Metrics (GEM) at Association for Computational Linguistics (ACL)*, 2021. [*Equal contribution]
- W6. **Aaron Mueller***, Yash Kumar Lal*. “**Sentence-Level Adaptation for Low-Resource Neural Machine Translation.**” In *Proceedings of the Workshop on Technologies for Machine Translation of Low-Resource Languages (LoResMT) at Machine Translation Summit (MTSummit)*, 2019. [*Equal contribution]

Invited Commentary & Editorials

- II. **Aaron Mueller**. “**Algorithm that gets ‘under the hood’ of AI models could effectively steer their responses.**” [Commentary on Beaglehole et al. (2026), “Toward universal steering and monitoring of AI models,” in *Science*.] In *Nature News and Views*, 2026.

Preprints & In Submission

- P1. Divya Appapogu, Freya Behrens, Yonatan Belinkov, **Aaron Mueller**. “**MAxBench: A Multinomial Concept Recovery Benchmark.**” arXiv preprint, 2026.

EDITOR

Workshop & Shared Task Proceedings

- EW1. Yonatan Belinkov, **Aaron Mueller**, Najoung Kim, Hosein Mohebbi, Hanjie Chen, Dana Arad, Gabriele Sarti. “**Proceedings of the 8th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP.**” 2025.
- EW2. Lucas Charpentier, Leshem Choshen, Ryan Cotterell, Mustafa Omer Gul, Michael Y. Hu, Jing Liu, Jaap Jumelet, Tal Linzen, **Aaron Mueller**, Candace Ross, Raj Sanjay Shah, Alex Warstadt, Ethan Gotlieb Wilcox, Adina Williams. “**Proceedings of the First BabyLM Workshop.**” 2025.
- EW3. Michael Y. Hu, **Aaron Mueller**, Candace Ross, Adina Williams, Tal Linzen, Chengxu Zhuang, Leshem Choshen, Ryan Cotterell, Alex Warstadt, Ethan Gotlieb Wilcox. “**The 2nd BabyLM Challenge at the 28th Conference on Computational Natural Language Learning.**” 2024.
- EW4. Yonatan Belinkov, Najoung Kim, Jaap Jumelet, Hosein Mohebbi, **Aaron Mueller**, Hanjie Chen. “**Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP.**” 2024.
- EW5. Alex Warstadt, **Aaron Mueller**, Leshem Choshen, Ethan Wilcox, Juan Ciro, Rafael Mosquera, Bhargavi Paranjabe, Adina Williams, Tal Linzen, Ryan Cotterell. “**Proceedings of the BabyLM Challenge at the 27th Conference on Computational Natural Language Learning.**” 2023.

INVITED TALKS AND PANELS

Concepts in Context: Understanding and Controlling Language Models from the Inside.

Human-Robot Interaction Seminar, Tufts University (Medford, MA). Oct. 9, 2026.

Does Interpretability Need to Be Mechanistic? [Joint talk with Tiago Pimentel.]

Keynote at the 2nd Big Picture Workshop, co-located with the 64th Annual Meeting of the Association for Computational Linguistics (ACL; San Diego, CA). Jul. 4, 2026.

Interpretability as an Engineering Tool for NLP.

Sheffield NLP Seminar Series, University of Sheffield (Sheffield, UK; virtual). Jun. 22, 2026.

Evaluating Methods for Locating and Controlling Concepts in Language Models.
10th Annual Center for Human-Compatible AI (CHAI) Workshop (Pacific Grove, CA). Jun. 7, 2026.

Locating and Steering Concepts in Language Models: A Causal Audit.
Language and Technology Lab (LTL), University of Cambridge (Cambridge, UK; virtual). May 7, 2026.

Understanding Incremental Processing in LLMs with Temporal Feature Analysis.
Text-as-Data Seminar Series, New York University (New York, NY). Feb. 5, 2026.

Building and Understanding Human-scale Language Models.
Natural Language & Information Processing (NLIP) Seminar Series, University of Cambridge (Cambridge, UK; virtual). Dec. 5, 2025.

Neurons, Directions, and Hulls: Rethinking the Geometry of Concepts in Large Language Models.
ML@FI Seminar Series, Flatiron Institute (New York, NY). Nov. 21, 2025.

Time- and Context-aware Interpretability.
Machine Learning and Friends Lunch, University of Massachusetts Amherst (Amherst, MA). Nov. 5, 2025.

Building a More Predictive Science of Language Model Behaviors with Interpretability.
Keynote at INTERPLAY workshop, co-located with the 2nd Conference on Language Modeling (COLM; Montréal, QC). Oct. 10, 2025.

Beyond Human Concepts: Evaluating and Applying Unsupervised Interpretability.
Keynote at the 2nd New England Mechanistic Interpretability Workshop (Boston, MA). Aug. 22, 2025.

The Quest for the Right Abstraction: A Causal Framework for Understanding Language Model Representations.
University of British Columbia Frontiers in NLP (Vancouver, BC). Jul. 17, 2025.

Panelist on reproducibility in AI research.
ACL Mentorship Session, NAACL (Albuquerque, NM). Apr. 30, 2025.

Causal Abstraction and Causal Emergence in Mechanistic Interpretability.
Bellairs Workshop on Causality (Barbados). Feb. 19, 2025.

Understanding and Controlling Language Models at the Feature Level.
Geva Lab, Tel Aviv University (Tel Aviv, Israel). Dec. 18, 2024.

Interpretability for Computational Linguistics: Understanding and Modifying How Models Make Decisions.
Language, Computation, and Cognition Lab, The Technion (Haifa, Israel). Nov. 24, 2024.

Mechanistically Controlling Language Models.
– Department of Computer Science, École polytechnique fédérale de Lausanne (EPFL; Lausanne, Switzerland). Jul. 4, 2024.
– Department of Computer Science, Saarland University (Saarbrücken, Germany). Jul. 2, 2024.

Bigger Is Not Always Better: The Benefits of Building and Understanding Sample-efficient Language Models.
Department of Computer Science, Maastricht University (Maastricht, Netherlands). Jun. 26, 2024.

Sparse Feature Circuits: Discovering and Editing Interpretable Causal Graphs in Language Models.
NLP Seminar, University of California, Santa Barbara (Santa Barbara, CA). Apr. 24, 2024.

Evaluating and Surgically Improving Generalization in Language Models.

- Responsible AI Seminar Series, Nokia Bell Labs (Cambridge, UK). Mar. 18, 2024.
- NLP Seminar, University of Pittsburgh (Pittsburgh, PA). Feb. 29, 2024.
- Deep Learning Superlab, Brown University (Providence, RI). Feb. 15, 2024.

Planting Trees in Language Models: Emergent Syntactic Behaviors and Mechanisms from Pre-training.

- Koller Lab, Saarland University (Saarbrücken, Germany). Feb. 7, 2023.
- NLP Seminar, Technion – Israel Institute of Technology (Haifa, Israel). Dec. 14, 2022.
- Bar-Ilan NLP Seminar, Bar-Ilan University (Ramat Gan, Israel). Dec. 13, 2022.

What Generalizations do Sequence-to-sequence Models Learn from Multilingual Text? Insights from Translation and Syntactic Transformations.

Multilingual Text Processing Group, National Research Council of Canada (Ottawa, ON). Mar. 4, 2022.

Syntactic Agreement in Neural Language Models: How Well and Where Do They Perform Subject-Verb Agreement?

Language & Understanding Group, Mila – Québec Artificial Intelligence Institute (Montréal, QC). Mar. 22, 2021.

Causal Mediation Analysis for Analyzing Neural Networks.

Fairness & Interpretability Research Talk Series, Google (New York, NY). Mar. 17, 2021.

Causal Analysis of Syntactic Agreement Mechanisms in Neural Language Models.

Center for Language & Speech Processing Seminar, Johns Hopkins University (Baltimore, MD). Feb. 12, 2021.

FUNDING	Aaron Mueller (PI). <i>Beyond Binary Variables: Defining, Locating, and Editing Multi-dimensional Features in Large Language Models.</i> National Science Foundation (award 2530728) and United States–Israel Binational Science Foundation (PI: Yonatan Belinkov). \$758,652.00 (\$471,529.00 to BU).	2025–2028
	Aaron Mueller (PI). <i>Interpretability Beyond Human Concepts.</i> Coefficient Giving.	2025–2027
FELLOWSHIPS AND AWARDS	Zuckerman Fellow, International	2023–2025
	Two-year postdoctoral fellowship. Supports research joint with an Israeli university and an American university. (\$126,000)	
	Microsoft Accelerate Foundation Models Research Award, International	2023
	Awarded for research on the capabilities of large language models. Provides OpenAI API credits and priority GPT-4 access. (\$10,000)	
	National Science Foundation Graduate Research Fellow, National	2018 - 2023
	Five-year graduate research fellowship. Provides three years of Ph.D. funding. (\$135,000)	
	Gaines Fellow, University of Kentucky	2016 - 2018
	Two-year fellowship. Requires the completion of a juried project, a thesis project, and a seminar in the humanities. (\$5,000)	
	Patterson Scholar, University of Kentucky	2014 - 2018
	Four-year scholarship covering tuition, educational materials, and room & board. Awarded to undergraduates who have earned National Merit semifinalist standing or higher. (\$86,000)	
	Goldwater Scholarship (Honorable Mention), <i>National</i>	2017
	Phi Beta Kappa, National	2017

	Raymond F. Betts Scholar , <i>University of Kentucky</i>	2017
	Awarded for thesis research. Used funds to design language technologies for low-resource dialects of French. (\$2,500)	
	Linguistics Research Award , <i>University of Kentucky</i>	2016
	Awarded to an undergraduate to facilitate a year-long research project in linguistics. (\$500)	
MEDIA APPEARANCES	“Kids outlearn AI—and we still don’t know why.” <i>MIT Technology Review</i>, Aug. 2026. “Forcing LLMs to be evil during training can make them nicer in the long run.” <i>MIT Technology Review</i>, Aug. 2025. “Apple, Microsoft Shrink AI Models to Improve Them.” <i>IEEE Spectrum</i>, Jun. 2024. “The Race to Make AI Smaller (and Smarter).” <i>The New York Times</i>, May 2023.	
MENTORING	Ph.D. advising	
	– Zhengyang Shan (BU). Research on developing interpretability frameworks for understanding and mitigating spurious correlations in LLMs.	2024–Pres.
	– Micah Benson (BU). Research on LLMs for social good.	2025–Pres.
	– Themistoklis Nikas (BU). Research on meta-cognitive capabilities in LLMs.	2025–Pres.
	– Divya Appapogu (BU). Research on high-dimensional concept discovery.	2025–Pres.
	– Nishkal Hundia (BU). Joint with Naomi Saphra. Research on long-context reasoning.	2026–Pres.
	– Vijay Murari Tiyyala (BU). Research on improving mixture-of-experts systems.	2026–Pres.
	Ad hoc Ph.D. mentoring	
	– Jannik Brinkmann (Northeastern). Joint with David Bau. Research on mechanistic interpretability for multilingual LLMs and protein LLMs. Resulted in a publication at NAACL [C20].	2024–Pres.
	– Aruna Sankaranarayanan (MIT). Joint with Dylan Hadfield-Menell. Research on natural and artificial grammar learning in language models. Resulted in a workshop publication [W4].	2023–2025
	– Juan Diego Rodriguez (UT Austin). Joint with Kanishka Misra. Research in how concepts are organized in language models. Resulted in a publication at NAACL [C21].	2023–2025
	– Dana Arad (Technion). Joint with Yonatan Belinkov. Research on principled applications of sparse autoencoders. Resulted in a publication at EMNLP [C16].	2024–2025
	– Tal Haklay (Technion). Joint with Yonatan Belinkov. Research on position-aware mechanistic interpretability methods. Resulted in a publication at ACL [C18].	2024–2025
	– Yaniv Nikankin (Technion). Joint with Yonatan Belinkov. Research on mechanistically understanding arithmetic in LLMs. Resulted in a publication at ICLR [C24].	2024–2025
	– Eric Todd (Northeastern). Joint with David Bau. Research on how functions are represented in neural language models. Resulted in a publication at ICLR [C26].	2023–2024
	Master’s students	
	– Hilton Raj (BU). Research on machine unlearning.	2026–Pres.
	– Angelina Sun (BU). Research on baselines for machine unlearning.	2026–Pres.
	– Jigar Milanbhai Kanakhara (BU). Research on how LLMs learn to be sycophantic.	2026–Pres.
	– Kerem Şahin (Northeastern). Joint with Chris Wendler. Research on the training dynamics of induction.	2025–2026
	– Swapnil Sharma (NYU). Research on evaluating summarization models.	2022–2023
	– Yash Kumar Lal (JHU). Resulted in a workshop publication [W6].	2018–2019

	Undergraduate researchers	
	– Jay Patel (BU). Research on emotion representations in LLMs.	2026–Pres.
	– Jacob Brinton (BU). Joint with Mark Crovella. Research on mechanistically understanding LLMs’ machine translation performance.	2025–2026
	– Keren Fuentes (Independent). Research on how LLMs represent user information.	2025–2026
	– Edward Sun (UCLA). SPAR mentee. Research on interpretable fine-tuning methods.	2026–Pres.
	– Yu Xia (NYU). Resulted in a publication at CoNLL [C34].	2021–2022
	– Matthew Finlayson (Harvard). Resulted in a publication at ACL [C38].	2020–2021
	BU qualifying exam committees	
	– Sahana Kowshik (Data Science)	2026
	– Yi Liu (Computer Science)	2026
	– Gabriel Franco (Computer Science)	2025
TEACHING	Boston University	
	<i>Instructor</i> , Advanced Natural Language Processing	Fall 2026
	<i>Instructor</i> , Introduction to Natural Language Processing	Spring 2026
	<i>Instructor</i> , Interpretable Machine Learning	Fall 2025
	Technion – Israel Institute of Technology	
	<i>Guest Lecture</i> , Introduction to Transformers	Spring 2025
	<i>Instructor</i> : Yonatan Belinkov	
	Massachusetts Institute of Technology	
	<i>Guest Lecture</i> , Quantitative Methods for NLP	Fall 2024
	<i>Instructor</i> : Jacob Andreas	
	New York University	
	<i>Guest Lecture</i> , Computational Linguistics & Cognitive Science	Spring 2023
	<i>Instructor</i> : Tal Linzen	
	Johns Hopkins University	
	<i>Teaching Assistant</i> , Machine Learning: AI System Design & Development	Spring 2020
	<i>Instructor</i> : Mathias Unberath	
SERVICE	Organizing committees	
	– BlackboxNLP 2026 (Co-located with EMNLP 2026)	
	– The 2026 BabyLM Workshop (Co-located with EMNLP 2026)	
	– The 3rd New England Mechanistic Interpretability Workshop (NEMI 2026)	
	– BlackboxNLP 2025 (Co-located with EMNLP 2025)	
	– The 2025 BabyLM Workshop (Co-located with EMNLP 2025)	
	– BlackboxNLP 2024 (Co-located with EMNLP 2024)	
	– The 2024 BabyLM Challenge	
	– The 2023 BabyLM Challenge	
	– The Inverse Scaling Prize (2022)	
	Grant panels	
	– Advisory Panel, U.S. National Science Foundation (NSF) (2026)	
	Conference chairing	
	– NAACL 2027 (Publicity Chair)	
	– EMNLP 2025 (Publicity Chair)	

Editorial responsibilities for scientific meetings

- EMNLP 2026 (Senior Area Chair for *Interpretability and Analysis of Models for NLP*)
- ACL 2026 (Area Chair for *Interpretability and Analysis of Models for NLP*)
- ICLR 2026 (Area Chair)
- EACL 2026 (Area Chair for *Interpretability and Analysis of Models for NLP*. **Outstanding Area Chair Award.**)
- EMNLP 2025 (Area Chair for *Interpretability and Analysis of Models for NLP*)
- COLM 2025 (Area Chair)
- ACL 2025 (Senior Area Chair for *Interpretability and Analysis of Models for NLP*)
- NAACL 2025 (Area Chair for *Interpretability and Analysis of Models for NLP*)
- EMNLP 2024 (Area Chair for *Interpretability and Analysis of Models for NLP*). **Outstanding Area Chair Award.**

Ad hoc journal reviewing

- Journal of Memory and Language (2025, 2024)
- Transactions of the Association for Computational Linguistics (2025)
- Computational Linguistics (2024)

Ad hoc grant reviewing

- Israel Science Foundation (2025)

Reviewing

- ICML (2026 [**Gold Reviewer Award**], 2025)
- ICLR (2025)
- NAACL (2024, 2021)
- ACL (2024, 2023, 2022, 2020)
- COLM (2024)
- NeurIPS (2024 [**Top Reviewer Award**])
- EACL (2024)
- FAccT (2024)
- EMNLP (2023, 2022, 2019)
- CoNLL (2023, 2022)
- TACL (2022)
- CSCW (2021)
- COLING (2020)

BU departmental service

- CS PhD Admissions Committee (2026-2027)
- Generative AI in Education Committee (2026–Present)
- IVC/ML Computing Lead and IT Liaison (2026–Present)

SKILLS

Programming

- Languages: Python, C++, HTML, CSS, Javascript, Bash
- Machine Learning Toolkits: PyTorch (incl. fairseq, sockeye), HuggingFace, vLLM, NLTK, Scikit-learn, numpy
- Version Control: DVCS (Git, Bitbucket)

Tools

- Linguistic tools: Praat, AntConc, QGIS, Audacity
- Annotation/survey tools: Prolific, Amazon Mechanical Turk, Qualtrics

NATURAL LANGUAGES

Native: English
Proficient: French
Beginner: Hebrew